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Communication control device and transmit/receive device for a subscriber station of a serial bus system and method for communication in a serial bus system

Abstract

A communication control device. The communication control device has a communication control module for generating a transmit signal for controlling communication of the subscriber station with other subscriber station(s) of the bus system which uses at least one first communication phase and one second communication phase for exchanging messages between subscriber stations of the bus system, and an interface module for encoding the transmit signal in at least one part of the second communication phase of a message into an encoded transmit signal, and for inserting additional data into at least one symbol of the encoded transmit signal. The interface module is configured to insert the additional data into the encoded transmit signal as at least one symbol of the encoded transmit signal serially with the information of the symbol of the transmit signal. The symbol duration of the transmit signal and of the encoded transmit signal is identical.

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Background/Summary**CROSS REFERENCE**

(1) The present application claims the benefit under 35 U.S.C. 119 of German Patent Application No. DE 10 2021 208 841.4 filed on Aug. 12, 2021, which is expressly incorporated herein by reference in its entirety.

FIELD

(2) The present invention relates to a communication control device and to a transmit/receive device for a subscriber station of a serial bus system and to a method for communication in a serial bus system operating at a high data rate, such as for example CAN XL.

BACKGROUND INFORMATION

(3) Serial bus systems are used for message or data transfer in engineering systems in which sensors and controllers are used. The serial bus system enables communication between the sensors and controllers of the engineering system. Such an engineering system is for example a vehicle or an industrial production plant or the like.

(4) There are various standards and data transfer protocols for data transfer in the bus system. A bus system in which data are transferred as messages using CAN FD to standard ISO 11898-1:2015 as the CAN protocol specification is frequently used for communication between sensors and controllers, for example in vehicles. The messages are transferred between the bus subscribers of

the bus system, such as a sensor, controller, transducer, etc. In CAN FD, transfer on the bus alternates between a slow operating mode in a first communication phase (arbitration phase) and a fast operating mode in a second communication phase (data phase). A data transfer rate of greater than 1 Mbit per second (1 Mbps) in the second communication phase is possible in a CAN FD bus system. Most manufacturers use CAN FD in the vehicle in the first step with an arbitration bit rate of 500 kbit/s and a data bit rate of 2 Mbit/s.

(5) In order to enable still higher data rates in the second communication phase, there are successor bus systems to CAN

(6) FD, such as for example CAN SIC and CAN XL. According to standard CiA601-4, a data rate of approximately 5 to 8 Mbit/s can be achieved in the second communication phase with CAN SIC. A data rate in the second communication phase of >10 Mbit/s is required in CAN XL, the standard (CiA610-3) for this system currently being defined by the organization CAN in Automation (CiA). In addition to pure data transfer via the CAN bus, CAN XL is also intended to support other functions, such as functional safety, data security and quality of service (QoS). These are fundamental characteristics which are required in self-driving vehicles.

(7) A serial signal for a message to be transmitted to the bus is generated in a subscriber station of the bus system using a communication control device which may include a protocol controller. The serial signal is forwarded to a transmit/receive device, which may also be denoted a transceiver.

The transmit/receive device generates from the serial signal at least one signal which is to be transferred on the bus of the bus system to other subscriber stations in the bus system.

Communication between the communication control device and the transmit/receive device of a subscriber station is necessary to ensure a correct changeover between the two different communication phases of CAN XL.

(8) At present, an additional terminal is required on both devices for such internal communication between the communication control device and the transmit/receive device which is not to be transmitted to the bus. However, this results in high costs, for example on the printed circuit board and in the microcontroller of the associated subscriber station. In addition, transfer of the other data may possibly lead to a reduction in the transferable data rate in the bus system.

SUMMARY

(9) It is an object of the present invention to provide a communication control device and a transmit/receive device for a subscriber station of a serial bus system and a method for communication in a serial bus system which solve the above-stated problems. In particular, an object is to provide a communication control device and a transmit/receive device for a subscriber station of a serial bus system and a method for communication in a serial bus system, in which communication between the communication control device and the transmit/receive device of a subscriber station can be achieved at low cost and with high error robustness and a high data rate in the bus system.

(10) The object may be achieved by a communication control device for a subscriber station of a serial bus system having the features of the present invention. In accordance with an example embodiment of the present invention, the communication control device has a communication control module for generating a transmit signal for controlling communication of the subscriber station with at least one other subscriber station of the bus system, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between subscriber stations of the bus system, and an interface module for encoding the transmit signal in at least one part of the second communication phase of a message into an encoded transmit signal, and for inserting additional data into at least one symbol of the encoded transmit signal, wherein the interface module is configured to insert the additional data into the encoded transmit signal as at least one symbol of the encoded transmit signal serially with the information of the symbol of the transmit signal, and wherein the symbol duration of the transmit signal and of the encoded transmit signal is identical.

(11) Using the communication control device, it is possible, without additional costly terminals between the communication control device and the transmit/receive device, to transfer additional data between the two devices. The communication control device is in this case configured to integrate the additional data into the transmit signal which is transmitted by the communication control device to the transmit/receive device for transmission to the bus. Nevertheless, integrating these additional data does not reduce the previous rapid data transfer in the serial bus system.

(12) The communication control device thus allows more data to be transferred between the communication control device and the transmit/receive device, despite the transfer frequency remaining the same, as for example defined in standard CiA610-3.

(13) Furthermore, using the communication control device, it is possible to retain arbitration as from CAN in one of the communication phases and nevertheless to achieve a further considerable increase in transfer rate over CAN or CAN FD.

(14) This can be achieved by using two communication phases with different bit rates and making the beginning of the second communication phase, in which the useful data are transferred at a higher bit rate than during arbitration, reliably identifiable for the transmit/receive device. The transmit/receive device can therefore reliably change over from a first communication phase into the second communication phase.

(15) As a consequence, a high bit rate and thus speed of transfer from transmitter to receiver is achievable. At the same time, however, high error robustness is ensured. This assists with achieving a net data rate of at least 10 Mbps.

(16) The method carried out by the communication control device may also be used when the bus system also includes at least one CAN subscriber station and/or at least one CAN FD subscriber station which transmit messages according to the CAN protocol and/or CAN FD protocol.

(17) Advantageous further example configurations of the communication control device are disclosed herein.

(18) According to one development of the present invention, the interface module is configured to carry out pulse-width modulation for encoding the transmit signal in at least one part of the second communication phase of the message.

(19) The interface module is possibly configured to subdivide the symbol duration of the transmit signal for the encoded transmit signal into n symbols for a first symbol of the transmit signal and into n symbols for a second symbol of the transmit signal, wherein n is a natural number greater than or equal to 1. The interface module may in this case be configured to insert the information of the symbol of the transmit signal into the first or last symbol of the $2 \cdot n$ symbols of the symbol duration of the encoded transmit signal. The interface module may alternatively be configured to insert the additional data between the first and last symbols of the $2 \cdot n$ symbols of the symbol duration of the encoded transmit signal.

(20) The communication control module is possibly configured to generate the transmit signal in the first communication phase of the message with symbols with a first bit time which is at least a factor of 10 greater than a second bit time of symbols which the communication control module generates in the transmit signal in the second communication phase of the message.

(21) The above-stated object may be additionally achieved by a transmit/receive device for a subscriber station of a serial bus system having the features of the present invention. In accordance with an example embodiment of the present invention, the transmit/receive device has a transmit module for transmitting a transmit signal to a bus of the bus system, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between subscriber stations of the bus system, an interface module for decoding the transmit signal in at least one part of the second communication phase of a message into a decoded transmit signal, and for extracting additional data from at least one symbol of the encoded transmit signal, and wherein the interface module is configured to extract the additional data from at least one symbol of the encoded transmit signal, which data have been inserted into the

encoded transmit signal serially with the information of the symbol of the transmit signal, and wherein the symbol duration of the transmit signal and of the decoded transmit signal is identical.

(22) According to one development of the present invention, the interface module is configured to carry out decoding of pulse-width modulation of the transmit signal in the at least one part of the second communication phase of the message.

(23) In accordance with an example embodiment of the present invention, in the transmit/receive device, the interface module may be configured to subdivide the symbol duration of the encoded transmit signal into n symbols for a first symbol of the encoded transmit signal and into n symbols for a second symbol of the encoded transmit signal, wherein n is a natural number greater than or equal to 1. The interface module is in this case configured to extract the information of the symbol of the transmit signal from the first or last symbol of the $2 \cdot n$ symbols of the symbol duration of the encoded transmit signal. The interface module is alternatively configured to extract the additional data between the first and last symbols of the $2 \cdot n$ symbols of the symbol duration of the encoded transmit signal.

(24) The transmit module may be configured to transmit the transmit signal to the bus as a differential signal.

(25) According to one option of the present invention, the transmit module is configured to generate the transmit signal in the first communication phase of the message and transmit it to the bus with a different physical layer than in the second communication phase of the message.

(26) In the above-described devices, the additional data may include at least one form of data of the following data, namely data for signaling to the transmit/receive device the operating mode into which the transmit/receive device has to switch its operating mode for the following communication phase, data for configuration of the transmit/receive device, data with a request to the transmit/receive device, data for signaling to the transmit/receive device the differential voltage with which the transmit/receive device is to operate for the bus, data for programming or flashing devices, data with a trigger for a predetermined wake-up pattern for a predetermined subscriber station of the bus system.

(27) It is possible for the first communication phase of the message to be provided in order to negotiate which of the subscriber stations of the bus system will receive at least temporarily exclusive, collision-free access to the bus in the following second communication phase.

(28) The above-described communication control device and the above-described transmit/receive device may be part of a subscriber station of a bus system which additionally comprises a bus and at least two subscriber stations which are connected together via the bus in such a manner that they can communicate serially with each other. At least one of the at least two subscriber stations in this case has an above-described communication control device and an above-described transmit/receive device.

(29) The above-stated object may be additionally achieved by a method for communication in a serial bus system in accordance with the present invention. In accordance with an example embodiment of the present invention, the method is carried out with a subscriber station for a bus system, in which at least one first communication phase and one second communication phase are used for exchanging messages between subscriber stations of the bus system, and wherein the method has the steps of generating, using a communication control module, a transmit signal for controlling communication of the subscriber station with at least one other subscriber station of the bus system, encoding, using an interface module, the transmit signal in at least one part of the second communication phase of a message into an encoded transmit signal, and inserting, using the interface module, additional data into at least one symbol of the encoded transmit signal, wherein the interface module inserts the additional data into the encoded transmit signal as at least one symbol of the encoded transmit signal serially with the information of the symbol of the transmit signal, and wherein the symbol duration of the transmit signal and of the encoded transmit signal is identical.

(30) The method offers the same advantages as are stated above in relation to the communication control device and/or the transmit/receive device.

(31) Further possible implementations of the present invention also include combinations which are not explicitly mentioned of features or embodiments as described above or hereinafter with regard to the exemplary embodiments. A person skilled in the art will in this case also add individual aspects as improvements or additions to the respective basic form of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention is described in greater detail below by way of exemplary embodiments and with reference to the figures.

(2) FIG. 1 shows a simplified block diagram of a bus system according to a first exemplary embodiment of the present invention.

(3) FIG. 2 shows a diagram illustrating the structure of messages which can be transmitted by subscriber stations of the bus system according to the first exemplary embodiment of the present invention.

(4) FIG. 3 shows a simplified schematic block diagram of a subscriber station of the bus system according to the first exemplary embodiment of the present invention.

(5) FIG. 4 to FIG. 7 each shows an example of a time plot respectively of signals or states or symbols at the subscriber station of FIG. 3, when the subscriber station is a transmitter of a message transmitted via a bus of the bus system.

(6) FIG. 8 to FIG. 11 each shows a time plot of encoding of the symbols L0 and L1 for a transmit signal by way of pulse-width modulation (PWM), in the event that the subscriber station is a transmitter of a message transmitted via a bus of the bus system.

(7) FIG. 12 to FIG. 13 each shows a time plot of encoding of the symbol L1 for a transmit signal by way of pulse-width modulation (PWM), in the event that the subscriber station is a receiver of a message transmitted via the bus of the bus system.

(8) FIG. 14 shows an example of the serial insertion of additional data into the symbols of FIG. 8 to FIG. 13 in the first exemplary embodiment.

(9) Unless otherwise stated, in the figures identical or functionally identical elements are provided with the same reference signs.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(10) FIG. 1 shows by way of example a bus system 1 which is in particular fundamentally configured for a CAN bus system, a CAN FD bus system, a CAN XL bus system, and/or modifications thereof, as described below. The bus system 1 may be used in a vehicle, in particular a motor vehicle, an aircraft, etc., or in a hospital etc.

(11) In FIG. 1, the bus system 1 has a plurality of subscriber stations 10, 20, 30, which are in each case connected to a bus 40 with a first bus wire 41 and a second bus wire 42. The bus wires 41, 42 may also be denoted CAN_H and CAN_L and serve for electrical signal transfer after injection of the dominant level or generation of recessive levels for a signal in the transmit state. Messages 45, 46 in the form of signals can be serially transferred between the individual subscriber stations 10, 20, 30 via the bus 40. The subscriber stations 10, 20, 30 are for example control devices, sensors, display devices, etc. of a motor vehicle.

(12) As FIG. 1 shows, the subscriber station 10 has a communication control device 11, a transmit/receive device 12, a first interface module 15 and a second interface module 16. In contrast, subscriber station 20 has a communication control device 21 and a transmit/receive device 22. Subscriber station 30 has a communication control device 31, a transmit/receive device 32, a first interface module 35 and a second interface module 36. The transmit/receive devices 12, 22, 32

of subscriber stations **10, 20, 30** are in each case directly connected to the bus **40**, even if this is not illustrated in FIG. 1.

(13) In each subscriber station **10, 20, 30**, the messages **45, 46** are encoded bit-wise in the form of frames via a TXD line and an RXD line between the respective communication control device **11, 21, 31** and the associated transmit/receive device **12, 22, 32**. This is described more precisely below.

(14) The communication control devices **11, 21, 31** in each case serve to control communication of the respective subscriber station **10, 20, 30** via the bus **40** with at least one other subscriber station of subscriber stations **10, 20, 30** which are connected to the bus **40**.

(15) The communication control devices **11, 31** create and read first messages **45** which are, for example, modified CAN messages **45** and are below also denoted CAN XL messages **45**. The CAN XL messages **45** are in this case structured on the basis of the CAN XL format, which is described in greater detail in relation to FIG. 2. The communication control devices **11, 31** may additionally be embodied in order, as required, to provide a CAN XL message **45** or a CAN FD message **46** for the transmit/receive devices **12, 32** or to receive it therefrom. The communication control devices **11, 31** thus create and read a first message **45** or second message **46**, wherein the first and second messages **45, 46** differ with regard to their data transfer standard, namely in this case CAN XL or CAN FD.

(16) The communication control device **21** may be embodied as a conventional CAN controller according to ISO 11898-1:2015, in particular in the manner of a CAN FD-tolerant classical CAN controller or a CAN FD controller. The communication control device **21** creates and reads second messages **46**, for example classical CAN messages or CAN FD messages **46**. The CAN FD messages **46** may comprise a number from 0 up to 64 data bytes which are additionally transferred at a distinctly higher data rate than a classical CAN message. In the latter case, the communication control device **21** is embodied as a conventional CAN FD controller.

(17) Apart from the differences described more precisely below, the transmit/receive devices **12, 32** can be embodied as a CAN XL transceiver and/or CAN SIC transceiver. The transmit/receive devices **12, 32** can additionally or alternatively be embodied as a conventional CAN FD transceiver. The transmit/receive device **22** can be embodied as a conventional CAN transceiver or CAN FD transceiver.

(18) In order to transmit messages **45, 47** using CAN XL and/or CAN SIC, tried and trusted characteristics are adopted which are responsible for the robustness and user friendliness of CAN and CAN FD, in particular a frame structure with identifier and arbitration using the conventional CSMA/CR method. The CSMA/CR method means that there must be “recessive” states on the bus **40** which can be overwritten by other subscriber stations **10, 20, 30** with dominant levels or dominant states on the bus **40**.

(19) Using the two subscriber stations **10, 30**, it is possible to form and then transfer messages **45** with the CAN XL format or CAN FD format and to receive such messages **45** with CAN FD transceivers and/or CAN SIC transceivers and/or CAN XL transceivers.

(20) FIG. 2 shows, for the message **45**, a frame **450**, in particular a CAN XL frame, as is provided by the communication control device **11** for the transmit/receive device **12** for transmission to the bus **40**. The communication control device **11** in this case creates the frame **450** in the present exemplary embodiment as compatible with CAN FD. The frame **450** is alternatively compatible with CAN SIC.

(21) According to FIG. 2, the frame **450** for CAN communication on the bus **40** is subdivided into different communication phases **451, 452**, namely an arbitration phase **451** (first communication phase) and a data phase **452** (second communication phase). The frame **450** has, after a start bit SOF, an arbitration field **453**, a control field **454**, a data field **455**, a checksum field **456**, and an end of frame field **457**.

(22) In the arbitration phase **451**, the subscriber stations **10, 20, 30** negotiate bit-by-bit, with the

assistance of an identifier (ID) for example using bits ID28 to ID18 in the arbitration field 453, which subscriber station 10, 20, 30 wishes to send the highest priority message 45, 46 and therefore receives exclusive access to the bus 40 of the bus system 1 for the next time for transmission in the subsequent data phase 452. The arbitration phase 451 uses a physical layer as in CAN and CAN-FD. The physical layer corresponds to the bit transfer layer or layer 1 of the conventional OSI (Open Systems Interconnection) model.

(23) One important point during phase 451 is that the conventional CSMA/CR method is used which permits simultaneous access of the subscriber stations 10, 20, 30 to the bus 40 without the higher priority message 45, 46 being destroyed. As a result, further bus subscriber stations 10, 20, 30 can relatively straightforwardly be added to the bus system 1, which is highly advantageous.

(24) The CSMA/CR method means that there must be “recessive” states on the bus 40 which can be overwritten by other subscriber stations 10, 20, 30 with dominant levels or dominant states on the bus 40. In the recessive state, high-resistance conditions prevail at the individual subscriber stations 10, 20, 30, which, in combination with bus switching parasitics, results in extended time constants. At present, this leads to a limitation of the maximum bit rate of today's CAN FD physical layer to approximately 2 megabits per second in real vehicle use.

(25) In addition to part of the control field 454, the useful data of the CAN XL frame 450 or the message 45 from the data field 455 and the checksum field 456 are transmitted in the data phase 452. The checksum field 456 may include a checksum regarding the data of the data phase 452 including stuff bits which are inserted by the transmitter of the message 45 after in each case a predetermined number of identical bits, in particular 10 identical bits, as an inverted bit. At the end of the data phase 452, operation is switched back to the arbitration phase 451.

(26) At least one acknowledge bit may be present in an end field in the end of frame phase 457. A sequence of 11 identical bits may moreover be present which indicate the end of the CAN XL frame 450. The at least one acknowledge bit can communicate whether or not a receiver has found an error in the received CAN XL frame 450 or message 45.

(27) A transmitter of the message 45 only begins transmitting bits of the data phase 452 to the bus 40 once the subscriber station 10 has won the arbitration as the transmitter and the subscriber station 10 as transmitter thus has exclusive transmission access to the bus 40 of the bus system 1.

(28) In the arbitration phase 451 as the first communication phase, subscriber stations 10, 30 thus in part, in particular up to (and including) the FDF bit, make use of a format from CAN/CAN FD according to ISO 11898-1:2015. However, in comparison with CAN or CAN FD, an increase in the net data transfer rate, in particular up to over 10 megabits per second, is possible in the data phase 452 as the second communication phase. Moreover, the volume of useful data per frame can be raised, in particular to approximately 2 kbytes or any other desired value.

(29) FIG. 3 shows the basic structure of the subscriber station 10 with the communication control device 11, the transmit/receive device 12, a system ASIC 14 and the interface modules 15, 16. The interface modules 15, 16 process encoding for changeover of the operating mode of the transmit/receive device 12 between operating modes of the two communication phases 451, 452. Furthermore, additional data 17 can be transmitted from the device 11 to the device 12 with the assistance of the interface modules 15, 16, as will be described more precisely below.

(30) The communication control device 11 is configured as a microcontroller or includes a microcontroller. The communication control device 11 processes signals of any desired application, for example a control device for a motor, a safety system for a machine or a vehicle, or other applications. The microcontroller is alternatively or additionally configured to acquire at least one physical quantity, such as temperature, pressure or other physical quantities, and to evaluate and/or forward the resultant information via the bus 40 to another subscriber station 10, 20, 30 of the bus system 1 of FIG. 1.

(31) In addition to the interface module 15, the communication control device 11 has a communication control module 111. The communication control module 111 is a protocol

controller which implements a communication protocol, for example the CAN protocol, in particular the protocol for CAN XL or CAN FD. The communication control module **111** is configured to output a signal TXD_PC to the first interface module **15**. The first interface module **15** has a memory block **151**, in which for example the data **17** are storable. The first interface module **15** additionally has an encoding block **152** which generates a transmit signal TXD from the signal TXD_PC. In the data phase **452**, the encoding block **152** carries out pulse-width modulation (PWM) of the signal TXD_PC and so generates a pulse-width-modulated transmit signal TXD for the data phase **452**. Additionally, in the data phase **452** the encoding block **152** can encode the additional data **17** into the signal TXD_PC in order to generate the transmit signal TXD. This is described more precisely below. The data **17** may be generated for example by the communication control device **11** and/or the communication control module **111** and stored in the memory block **151**.

(32) In contrast, in the arbitration phase **451**, the encoding block **152** does not carry out any encoding of the signal TXD_PC. In the arbitration phase **451**, signal TXD_TC=transmit signal TXD therefore applies.

(33) The first interface module **15** outputs the transmit signal TXD to the terminal TxD of the device **11**. The communication control module **111** is additionally configured to receive a signal RXD from the terminal RxD of the device **11**.

(34) In the example of FIG. 3, the transmit/receive device **12** is integrated into the system ASIC **14** (ASIC=application-specific integrated circuit). Alternatively, however, the transmit/receive device **12** may be a separate device. The system ASIC **14** may alternatively be a system basis chip (SBC) on which a plurality of functions required for an electronics assembly of the subscriber station **10** are combined. The system ASIC **14** may inter alia include the transmit/receive device **12** and a power supply device (not shown) which supplies the transmit/receive device **12** with electrical energy. The power supply device usually supplies a voltage CAN Supply of 5 V. If required, however, the power supply device can supply another voltage with a different value and/or be configured as a power source. The transmit/receive device **12** additionally has a transmit module **121** and a receive module **122**.

(35) The first interface module **15** of the communication control device **11** is of symmetrical construction to the second interface module **16** of the transmit/receive device **12**. The first interface module **15** may also be denoted PC interface module **15**. The second interface module **16** may also be denoted TC interface module **16**. The first interface module **15** transmits the transmit signal TXD from the terminal TxD of the communication control device **11** to the terminal TxD of the transmit/receive device **12**. The transmit signal TXD is thus transferred between the interface modules **15**, **16**.

(36) The interface module **16** has a memory block **161** and a decoding block **162**. In the data phase **452**, the decoding block **162** carries out decoding of the signal TXD in order to eliminate the pulse-width modulation. In addition, the decoding block **162** extracts the data **17** from the transmit signal TXD. The decoding block **162** can store the data **17** in particular in the memory block **161**. The data **17** are usable in the transmit/receive device **12** and/or the system ASIC **14**. The decoding block **162** thus outputs the pulse-width-demodulated transmit signal TXD_TC to the transmit module **121**. The transmit signal TXD_TC is processed by the transmit module **121** for transmission to the bus **40**.

(37) In contrast, in the arbitration phase **451**, the decoding block **152** does not have to carry out any decoding of pulse-width modulation of the signal TXD. In principle, however, the signal TXD can be NRZ-encoded. In this case, NRZ=non-return-to-zero=encoding of the signal onto two different conduction states, usually voltage states, with constant levels during a symbol. In the arbitration phase **451**, transmit signal TXD=signal TXD_TC therefore applies.

(38) The subscriber station **30** is of similar construction, as shown in FIG. 3, except that module **35** is not integrated into the communication control device **31** but instead provided separately from the

communication control device **31** and the transmit/receive device **32**. Subscriber station **30** and module **35** are therefore not separately described. The functions described below of the first interface module **15** are identically present in the first interface module **35**. The functions described below of the second interface module **16** are identically present in the second interface module **36**. (39) It is alternatively or additionally possible for the module **16** not to be integrated into the transmit/receive device **12** but instead to be provided separately from the communication control device **11** and the transmit/receive device **12**.

(40) As shown in FIG. 3, the transmit/receive device **12** is connected to the bus **40**, or more precisely its first bus wire **41** for CAN_H and its second bus wire **42** for CAN_L. During operation of the bus system **1**, the transmit/receive device **12**, or more precisely its transmit module **121**, converts the transmit signal TXD_TC into corresponding signals CAN_H and CAN_L with voltage levels U for the bus wires **41**, **42**. The transmit module **121** transmits these signals CAN_H and CAN_L over time t to the bus **40**, as shown in FIG. 4. The receive module **122** evaluates signals CAN_H and CAN_L received from the bus **40** and/or their differential signal $V_{DIFF} = CAN_H - CAN_L$ shown in FIG. 5 and forms a receive signal RXD. Ideally, the receive signal RXD is identical to the transmit signal TXD_TC. The receive module **122** outputs the receive signal RXD at the terminal RxD of the device **12**.

(41) Despite the signals for the transmit/receive device **12** being here denoted CAN_H and CAN_L, they should not be understood in relation to message **45** as signals CAN-XL_H and CAN-_L which, in the data phase **452**, deviate from the conventional signals CAN_H and CAN_L in at least one feature, in particular with regard to the formation of the bus states for the various data states of the signal TXD and/or with regard to the voltage or the physical layer and/or the bit rate.

(42) FIG. 4 shows on the left-hand side that, in the arbitration phase **451**, subscriber stations **10**, **20**, **30** transmit signals CAN_H, CAN_L to the bus **40** which alternately have at least one dominant state **401** or at least one recessive state **402**. After arbitration in the arbitration phase **451**, one of subscriber stations **10**, **20**, **30** is established as the winner. Assuming that subscriber station **10** has won the arbitration, the transmit/receive device **12** of subscriber station **10** changes over its physical layer at the end of the arbitration phase **451** from a first operating mode (SLOW) into a second operating mode (FAST_TX) since subscriber station **10** is the transmitter of the message **45** in the data phase **452**. In the data phase **452** or in the second operating mode (FAST_TX), the transmit module **121** then, as a function of a transmit signal TXD, generates symbols L0 or L1 in succession and thus serially in the signals CAN_H, CAN_L on the bus **40**. The frequency of the signals CAN_H, CAN_L in the data phase **452** can be increased, as shown on the right-hand side in FIG. 4. Accordingly, the symbol duration t_{pr0} in the arbitration phase **451** (first communication phase) is greater than the symbol duration t_{pr} in the data phase **452** (second communication phase). The net data transfer rate is thus increased in the data phase **452** in comparison with the arbitration phase **451**. On the other hand, the transmit/receive device **32** of subscriber station **30** changes over its physical layer at the end of the arbitration phase **451** from the first operating mode (SLOW) into a third operating mode (FAST_RX) since in the data phase **452** subscriber station **30** is only a receiver, i.e. not a transmitter, of the frame **450**. After the end of the arbitration phase **451**, all the transmit/receive devices **12**, **32** of subscriber stations **10**, **30** change over their operating mode into the first operating mode (SLOW). All the transmit/receive devices **12**, **32** thus also change over their physical layer.

(43) The changeover of the operating modes of the transmit/receive devices **12**, **32** between the operating mode for the arbitration phase **451** and the operating modes FAST_TX or FAST_RX is controlled via the encoding of the data in the signal TXD on the TXD line between the terminals TxD of the devices **11**, **12**. As mentioned above, encoding in the PC interface module **15** of the device **11** is performed using the data **17**. This function of the PC interface module **15** may also be denoted PWM encoder. Decoding of the data **17** from the signal TXD is performed in the TC interface module **16** of the device **12**. This function of the TC interface module **16** may also be

denoted PWM decoder.

(44) According to FIG. 4 and FIG. 5, a differential signal $V_{DIFF} = CAN_H - CAN_L$ with values of $V_{DIFF} = 2\text{ V}$ for dominant states **401** and $V_{DIFF} = 0\text{ V}$ for recessive states **402** is ideally obtained in the arbitration phase **451** on the bus **40**. This is shown on the left-hand side in FIG. 4 and FIG. 5. In contrast, a differential signal $V_{DIFF} = CAN_H - CAN_L$ with states L0, L1 is obtained in the data phase **452** on the bus **40**, as shown on the right-hand side in FIG. 4 and FIG. 5.

(45) FIG. 6 shows for the data phase **452** an example of the profile over time t of a transmit signal TXD_PC generated by the communication control module **111**. The transmit signal TXD_PC has both symbols HI (high) and LW (low). The transmit signal TXD_PC is a digital signal.

(46) State L0 has a value $V_{DIFF} = 1\text{ V}$. State L1 has a value $V_{DIFF} = -1\text{ V}$. The receive module **122** can in each case distinguish states **401**, **402** and L0, L1 with the receive thresholds which are located in the ranges TH_T1, TH_T2, TH_T3. The receive module **122** in this case uses at least the receive threshold T1 of for example 0.7 V in the arbitration phase **451**. The receive module **122** uses the receive threshold T2 of for example -0.35 V for example in the arbitration phase **451**, but optionally also in the data phase **452**. The receive threshold T3 of for example 0.0 V is used in the data phase **452**. On changeover between the first to third operating modes (SLOW, FAST_TX, FAST_RX), which have been described above, the receive module **122** in each case changes over the receive thresholds T1, T2, T3.

(47) With the exception of an idle or standby state, the transmit/receive device **12** may be set up such that the device **12** with its receiver is always listening in normal operation for a transfer of data or messages **45**, **46** on the bus **40**, specifically irrespective of whether subscriber station **10** is or is not the transmitter of the message **45**. The transmit/receive device **12** forms the receive signal RXD from the signals CAN_H and CAN_L received from the bus **40** and forwards it to the communication control device **11**, as described above.

(48) FIG. 8 to FIG. 11 show, in the event that subscriber station **10** is the transmitter of frame **450** in the data phase **452**, in each case different examples of the encoding which the encoding block **152** carries out over time t for a symbol with a symbol duration $t_{pr} = 2 \cdot 1/2 t_{pr}$ in order to convert the signal TXD_PC into the transmit signal TXD. In this case, the subscriber station **10** is switched into the operating mode FAST_TX in the data phase **452**.

(49) According to FIG. 8 to FIG. 11, the resultant pulse-width-modulated signal TXD changes during the symbol duration t_{pr} in each case between the levels high HI and low LW. The symbol duration t_{pr} begins at a point in time t_1 , has half elapsed at a point in time t_2 and finishes at a point in time t_3 . At points in time t_1 , t_3 , the signal TXD in each case has approximately the value 0.5 HI . A level of approximately 50% of the level HI is thus in each case present at the beginning and end of the symbol duration t_{pr} .

(50) With CAN XL, a pulse-width modulation (PWM) is used in the second communication phase (data phase) **452** for the symbols L0, L1 in the transmit signal TXD_PC . Standard CiA610-3 in this cases defines minimum and maximum pulse lengths of the symbols L0, L1 in the signal TXD between the communication control device **11** and the transmit/receive device **12** for decoding symbols L0, L1 in the transmit/receive device **12** with a time $t_{decode} \leq 5\text{ ns}$, as shown in FIG. 8 to FIG. 11. Standard CiA610-3 additionally defines that in the case of a subscriber station **10**, **30**, which in ongoing transfer of a message **45** on the bus **40** is not a transmitter but only a receiver of the message **45**, the communication control device **11**, **31** is exclusively permitted to transmit the symbol L1 in the transmit signal TXD_PC . For changeover of the communication phases **451**, **452**, **451**, it is likewise possible to use pulse-width modulation with the symbols L0, L1 for communication between the communication control device **11**, **31** and the transmit/receive device **12**, **32** of a subscriber station **10**, **30**.

(51) FIG. 8 and FIG. 9 show in each case different examples of the encoding of the state or symbol L0 in the transmit signal TXD. FIG. 8 in this case shows encoding in which the transmit signal TXD has a symbol L0_ma. The symbol L0_ma has the maximum admissible duration for the level

high HI and thus the minimum admissible duration for the level low LW. In contrast, FIG. 9 shows encoding in which the transmit signal TXD has a symbol L0_mi. The symbol L0_mi has the minimum admissible duration for the level high HI and thus the maximum admissible duration for the level low LW. In the case of both symbols L0_ma, L0_mi, the duration for the level high HI is <than half the symbol duration t_{pr} . In addition, both symbols L0_ma, L0_mi begin with the level high HI and end with the level low LW.

(52) FIG. 10 and FIG. 11 show in each case different examples of the encoding of the state or symbol L1 in the transmit signal TXD. FIG. 10 in this case shows encoding in which the transmit signal TXD has a symbol L1_mi. The symbol L1_mi has the minimum admissible duration for the level high HI and thus the maximum admissible duration for the level low LW. In contrast, FIG. 11 shows encoding in which the transmit signal TXD has a symbol L1_ma. The symbol L1_ma has the maximum admissible duration for the level high HI and thus the minimum admissible duration for the level low LW. In the case of both symbols L1_ma, L1_mi, the duration for the level high HI is greater than half the symbol duration t_{pr} . In addition, both symbols L1_mi, L1_ma begin with the level high HI and end with the level low LW.

(53) According to FIG. 8 to FIG. 11, each of the symbols L0_ma, L0_mi, L1_mi, L1_ma of the PWM-encoded transmit signal TXD is decoded by the decoding block 162 with decoding time t_{dc} , or more precisely is acquired with a temporal resolution of t_{dc} and thus converted into the transmit signal TXD_TC. The decoding time t_{dc} is set to a predetermined value. For example, $t_{dc} \leq 5$ ns applies, as mentioned above for CAN XL. The decoding time t_{dc} is displaceable in the symbol duration t_{pr} . The decoding time t_{dc} in this case begins with the point in time t1 or the decoding time t_{dc} ends with the point in time t3. Alternatively, the decoding time t_{dc} begins or ends with the point in time t2. The decoding time t_{dc} thus in each case begins or ends at a level of approximately 50% of the level HI. The level which the decoding block 162 identifies in the decoding time t_{dc} is generated as the level of the transmit signal TXD_TC.

(54) FIG. 12 and FIG. 13 show signal profiles in the event that the subscriber station 10 is only the receiver of the frame 450 in the data phase 452. FIG. 12 and FIG. 13 in each case show different examples of the encoding which the encoding block 152 carries out over time t for a symbol with the symbol duration $t_{pr} = 2 \cdot 1/2 t_{pr}$ in order to convert the signal TXD_PC into the transmit signal TXD. In this case, the subscriber station 10 is switched into the operating mode FAST_RX in data phase 452. In the CAN XL operating mode FAST_RX, the communication module 111 is only permitted to generate the transmit signal TXD_PC in such a way that only symbols L1 are transmitted to the bus 40, as already mentioned.

(55) According to FIG. 12 and FIG. 13, the resultant pulse-width-modulated signal TXD changes during the symbol duration t_{pr} in each case between the levels high HI and low LW, as described above for FIG. 10 and FIG. 11.

(56) FIG. 12 and FIG. 13 in each case show, in contrast to FIG. 10 and FIG. 11, different examples of the encoding of the state L1_ma_RX, L1_mi_RX in the transmit signal TXD. FIG. 12 in this case shows encoding in which the transmit signal TXD has a symbol L1_ma_RX. The symbol L1_ma_RX has the maximum admissible duration for the level low LW and thus the minimum admissible duration for the level high HI. In contrast, FIG. 13 shows encoding in which the transmit signal TXD has a symbol L1_mi_RX. The symbol L1_mi_RX has the minimum admissible duration for the level low LW and thus the maximum admissible duration for the level high HI. In the case of both symbols L1_ma_RX, L1_mi_RX, the duration for the level high HI is >than half the symbol duration t_{pr} . In addition, both symbols L1_ma_RX, L1_mi_RX begin with the level low LW and end with the level high HI.

(57) It is configured in the encoding block 152 for the operating mode FAST_TX with which of the symbols of FIG. 8 or FIG. 9 a symbol L0 is to be encoded in the transmit signal TXD_PC and with which of the symbols of FIG. 10 or FIG. 11 a symbol L1 is to be encoded in the transmit signal TXD_PC. This also gives rise to the configuration for the operating mode FAST_RX for generating

the symbols of FIG. 12 or FIG. 13. The symbol duration t_{pr} is additionally defined in the encoding block 152. The configuration of the encoding block 152 also gives rise to the configuration of the decoding block 162, in particular also when the decoding time t_{dc} begins and/or ends.

(58) In order to encode the additional data 17 into the symbols of FIG. 8 to FIG. 13, the encoding block 152 proceeds as described on the basis of FIG. 14.

(59) According to FIG. 14, the encoding block 152 can modulate in a symbol duration t_{pr} a level L0 of the transmit signal TXD_PC with $2 \cdot n$ symbols for the pulse-width-modulated transmit signal TXD, wherein n is a natural number greater than or equal to 1. In addition, the encoding block 152 can modulate a level L1 of the transmit signal TXD_PC with $2 \cdot n$ symbols for the pulse-width-modulated transmit signal TXD. In the example of FIG. 14, the symbols L0-n, L0-1, L0, L0+1, L0+n, L1-n, L1-1, L1, L1+1, L1+n are available in a symbol duration t_{pr} . Of these, a minimum of only the first and last symbols L0-n, L1+n are necessary for identifying the level L0 or L1 which is to be transferred by the transmit signal TXD_PC into the pulse-width-modulated transmit signal TXD. The other symbols can be used for encoding the data 17.

(60) FIG. 14 shows an example configuration of the encoding block 152 in the operating mode FAST_TX, in which a configuration according to FIG. 9 is selected for a symbol L0 and a configuration according to FIG. 11 for a symbol L1. The data 17 can therefore be inserted between the first and last symbol L0-n, L1+n. Very generally, the encoding block 152 can insert the additional data 17 into the encoded transmit signal TXD as at least one of the symbols L0-n, L1+n of the encoded transmit signal TXD serially with the information of the symbol L0, L1 of the transmit signal TXD_PC.

(61) On decoding, the decoding block 162 can then read back out from first and last symbols L0-n, L1+n whether the symbol L0 or L1 should be introduced into the signal TXD_TC. In addition, on decoding the decoding block 162 can extract the data 17 and for example save them in the memory block 161. A similar situation applies to the configuration of the encoding block 152 in the operating mode FAST_RX.

(62) If, for example, the data 17 contain the operating mode into which the transmit/receive device 12 is to be switched, the data 17 are used for switching the operating mode of the transmit/receive device 12.

(63) As a result, the data rate of the transmit signal TXD may be increased for at least one symbol of the signal of $(L0/L1)=1/t_{pr}$ to L0 modulated with $2 \cdot n$ symbols, L1 modulated with $2 \cdot n$ symbols $= 2 \cdot n / t_{pr}$.

(64) The value n for configuring the encoding block 152 is selectable for example as a function of the bit rate which is used. In CAN-XL with 5 Mbit/s in operating modes FAST-TX, FAST-RX, the symbol duration amounts to $t_{pr}=200$ ns according to standard CiA610-3. In CAN-XL with 20 Mbit/s in operating modes FAST-TX, FAST-RX, the symbol duration amounts to $t_{pr}=50$ ns according to standard CiA610-3. As already mentioned above, the defined accuracy of PWM decoding is defined, independently of bit rate, at a maximum of 5 ns. n can thus be selected low at higher bit rates and high at low bit rates.

(65) According to one specific variant embodiment, $n=1$ applies. All the bit rates which are specified in standard CiA610-3 can be achieved therewith.

(66) The data 17 can be inserted only in some of the possible symbols between the first and last symbols L0-n, L1+n. The data 17 are inserted in at least one symbol between the first and last symbols L0-n, L1+n. In particular, the data 17 can only be inserted into the part of the possible symbols for L0 modulated with $2 \cdot n$ symbols or only into the part of the possible symbols for L1 modulated with $2 \cdot n$ symbols.

(67) If required, the data 17 can be distributed between more than one symbol duration t_{pr} . This is advantageous if a relatively large volume of data 17 is to be transferred and/or the bit rate is rather high.

(68) Modulation by the encoding block **152** may proceed in each pulse width modulation period (PWM period) or only as required, in particular only in at least one pulse width modulation period (PWM period).

(69) The method can be applied for transmitting subscriber stations **10**, **30** (Fast-TX subscriber stations **10**, **30**) and/or for receiving subscriber stations **10**, **30** (Fast-RX subscriber stations **10**, **30**).

(70) The described structure of the subscriber stations **10**, **30** thus provides a robust and simple possible way to transfer additional data **17** between the communication control device **11** and the transmit/receive device **12** without additional costly terminals. Data rates are also not reduced in this case but, on transfer of data during the data phase **452** of a frame **450**, are instead even raised in comparison with conventional techniques.

(71) According to one modification of the first exemplary embodiment, it is possible for the encoding block **152** to carry out, instead of pulse-width modulation, another encoding of the transmit signal TXD_PC. Any encoding which, in the encoded transmit signal TXD, serially inserts the information of the symbols L0, L1 of the transmit signal TXD_PC within the previous symbol duration t_{pr} is suitable for this purpose.

(72) According to a second exemplary embodiment, the encoding block **152** is configured to carry out configuration of the transmit/receive device **12** with the data **17**. In this way, configuration of the transmit/receive device **12** may proceed so to speak in concealed manner in the TXD data stream.

(73) For example, in the present exemplary embodiment, the encoding block **152** is configured in a CAN XL frame **450** to transmit the last two 0 bits (FPC1, FCP0) with the symbol L0-1 instead of with a symbol L0. The transmit/receive device **12** then identifies, in particular by means of the decoding block **162**, that the transmit/receive device **12** is to enter the standby state shortly after being switched back from the operating mode FAST-TX into the operating mode SLOW of the arbitration phase **451**.

(74) This configuration of the subscriber station **10** may lead to economies on transmit/receive device **12** (transceiver) terminals. For example, the separately present terminal STB (standby terminal) can be omitted. This is highly advantageous in the case of a separate transmit/receive device **12** with eight terminals (8 pin standalone transceiver), since separate grounds for the transmit module **121** (transmitter) and the terminal RxD can very straightforwardly be enabled in this way. Emissions limit values can consequently be observed at the required CAN-XL bit rates.

(75) According to a third exemplary embodiment, the encoding block **152** is configured to transmit at least one request to the transmit/receive device **12** with the data **17**.

(76) For example, as a result of the request in the data **17**, the transmit/receive device **12** and/or the system ASIC **14** can acquire at least one physical quantity. The associated physical data may in particular be provided to the encoding block **152** and/or the communication control module **111** and/or the device **11**. The physical data may for example be transmitted to the encoding block **152** via a separate message. Such physical data for the data **17** may for example be data relating to the corrosion of electrical connection contacts on the bus lines **41**, **42**. Depending on corrosion, it may be necessary to take steps on the device **11** to modify the differential voltage VDIFF of the transmit signal TxD. The quality of data transfer in the bus system **1** can in this way be ensured (quality of service (QoS)).

(77) According to a fourth exemplary embodiment, the encoding block **152** is configured to communicate to the transmit/receive device **12** with the data **17** the differential voltage VDIFF with which the transmit/receive device **12** is to operate. For example, the symbol L0 may mean that, in the operating modes FAST-TX, FAST-RX, the transmit/receive device **12** is to operate with the differential voltages VDIFF of FIG. 5. The symbol L0-1 may mean that the 0 level is to be transmitted with a smaller differential voltage VDIFF than in the example of FIG. 5. The same applies for the symbol L1. The symbol L1+1 may accordingly mean that the 1 level is to be transmitted with a larger differential voltage VDIFF than in the example of FIG. 5.

(78) According to a fifth exemplary embodiment, the encoding block **152** is configured to communicate or initiate or carry out programming/flashing of devices with the data **17**. Each electronic component which contains the transmit/receive device **12** (CAN-XL transceiver module), including the terminal TxD, can be programmed via the interface modules **15**, **16**. For example, the data **17** can specify the following configuration, namely a bit rate configuration for 10 Mbit/s, whereby $t_{pr_TxD}=100$ ns applies. In addition, a conservative/robust configuration $n=2$ may be specified, resulting in a data rate of the transmit signal TxD as $2 \cdot n / t_{pr} = 4 / 100 \text{ ns} = 25$ Mbaud.

(79) According to a sixth exemplary embodiment, the encoding block **152** is configured to initiate or carry out triggering for selective wake-up bus signals with the data **17**. The encoding block **152** can thus trigger the transmit/receive device **12** to transmit a specific wake-up pattern to the bus **40**, in order for example to reawaken just one individual subscriber station **10**, **20**, **30** of the bus system **1** after said subscriber has not been required for a certain period and has therefore been put to sleep. For example, selective wake-up is possible in CAN-XL by means of a pulse with a voltage level of -1 V.

(80) All the above-described configurations of the devices **11**, **12**, **31**, **32**, of the modules **15**, **16**, **35**, **36**, of the subscriber stations **10**, **20**, **30**, of the bus system **1** and of the method carried out therein may be used individually or in any possible combinations. In particular, all the features of above-described exemplary embodiments and/or the modifications thereof may be combined as desired. In particular, the following modifications are additionally or alternatively possible.

(81) Although the present invention has been described above taking the CAN bus system by way of example, the present invention can be used in any communication network and/or communication method in which two different communication phases are used in which there are differences in the bus states which are generated for the different communication phases. In particular, the above-described principle of the present invention is usable in interfaces which require for different communication phases a changeover signal from a protocol controller or module **111** and/or in this case require data exchange between devices **11**, **12**.

(82) The above-described bus system **1** according to the exemplary embodiments is described on the basis of a bus system based on the CAN protocol. The bus system **1** according to the exemplary embodiments may however alternatively be another kind of communication network in which data are serially transferable at two different bit rates. It is advantageous, but not a necessary prerequisite, that exclusive, collision-free access of a subscriber station **10**, **20**, **30** to a common channel is ensured in the bus system **1** at least for specific time intervals.

(83) At least one form of data **17**, which is described in the preceding exemplary embodiments, can be integrated in a frame **45** between the devices **11**, **12** or **31**, **32**. Any desired combination of the data **17** which are described in the preceding exemplary embodiments is thus possible. In addition, any desired arrangement of the data **17** in the data phase **452** of a message **45** or of a frame **450** is possible.

(84) Any desired number and arrangement of the subscriber stations **10**, **20**, **30** in the bus system **1** of the exemplary embodiments is possible. In particular, subscriber station **20** can be omitted from the bus system **1**. It is possible for one or more of subscriber stations **10** or **30** to be present in the bus system **1**. It is possible for all the subscriber stations in the bus system **1** to be identically configured, i.e. for only subscriber station **10** or only subscriber station **30** to be present.

Claims

1. A communication control device for a subscriber station of a serial bus system that includes a plurality of subscriber stations, the subscriber station being one of the plurality of subscriber stations, comprising: a communication control module configured to generate a transmit signal for a transceiver of the subscriber station that communicates with at least one other subscriber station

of the plurality of subscriber stations of the bus system, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between the plurality of subscriber stations of the bus system, wherein the transmit signal includes data of one of the messages to be transmitted by the transceiver; and an interface module configured to: encode the transmit signal in at least one part of the second communication phase of the one of the messages into an encoded transmit signal that includes at least one symbol; and insert additional data, that controls a behavior of the transceiver, into the at least one symbol of the encoded transmit signal, such that the additional data is interleaved serially with the data of the at least one symbol of the one of the messages of the transmit signal; and wherein a symbol duration of the transmit signal and of the encoded transmit signal is identical.

2. The communication control device as recited in claim 1, wherein the interface module is configured to carry out pulse-width modulation for encoding the transmit signal in at least one part of the second communication phase of the message.

3. A communication control device for a subscriber station of a serial bus system, the communication control device comprising: a communication control module configured to generate a transmit signal for controlling communication of the subscriber station with at least one other subscriber station of the bus system, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between subscriber stations of the bus system; and an interface module configured to encode the transmit signal in at least one part of the second communication phase of a message into an encoded transmit signal, and to insert additional data into at least one symbol of the encoded transmit signal; wherein: the interface module is configured to insert the additional data into the encoded transmit signal as at least one symbol of the encoded transmit signal serially with information of the symbol of the transmit signal; a symbol duration of the transmit signal and of the encoded transmit signal is identical; and the interface module is configured to subdivide the symbol duration of the transmit signal for the encoded transmit signal into n symbols for a first symbol of the transmit signal and into n symbols for a second symbol of the transmit signal, wherein n is a natural number greater than or equal to 1.

4. The communication control device as recited in claim 3, wherein the interface module is configured to insert the information of the symbol of the transmit signal into the first or last symbol of the $2*n$ symbols of the symbol duration of the encoded transmit signal.

5. The communication control device as recited in claim 3, wherein the interface module is configured to insert the additional data between the first and last symbols of the $2*n$ symbols of the symbol duration of the encoded transmit signal.

6. A communication control device for a subscriber station of a serial bus system, the communication control device comprising: a communication control module configured to generate a transmit signal for controlling communication of the subscriber station with at least one other subscriber station of the bus system, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between subscriber stations of the bus system; and an interface module configured to encode the transmit signal in at least one part of the second communication phase of a message into an encoded transmit signal, and to insert additional data into at least one symbol of the encoded transmit signal; wherein: the interface module is configured to insert the additional data into the encoded transmit signal as at least one symbol of the encoded transmit signal serially with information of the symbol of the transmit signal; a symbol duration of the transmit signal and of the encoded transmit signal is identical; and the communication control module is configured to generate the transmit signal in the first communication phase of the message with symbols with a first bit time which is at least a factor of 10 greater than a second bit time of symbols which the communication control module generates in the transmit signal in the second communication phase of the message.

7. A transmit/receive device of a subscriber station of a serial bus system that includes a plurality of

subscriber stations, the subscriber station being one of the plurality of subscriber stations, wherein the bus system is configured to use a first communication phase and a second communication phase for exchanging messages between the plurality of subscriber stations of the bus system, the transmit/receive device comprising: a transmit module configured to transmit, to another one of the subscriber stations and via a bus of the bus system, data of one of the messages, the data having been encoded, from a transmit signal, with at least one symbol in an encoded transmit signal; and an interface module configured to decode the encoded transmit signal in at least one part of the second communication phase of the one of the messages to extract additional data from the at least one symbol of the encoded transmit signal; wherein: the additional data had been inserted into the at least one symbol of the encoded transmit signal to be interleaved serially with the data of the at least one symbol of the one of the messages; a behavior of the transmit module is controlled according to the extracted additional data; and a symbol duration of the transmit signal and of the encoded transmit signal is identical.

8. The transmit/receive device as recited in claim 7, wherein the interface module is configured to carry out decoding of pulse-width modulation of the transmit signal in the at least one part of the second communication phase of the message.

9. The transmit/receive device as recited in claim 7, wherein the transmit module is configured to transmit the data of the one of the messages as a differential signal to the bus.

10. The transmit/receive device as recited in claim 7, wherein the transmit module is configured to generate a transmit signal in the first communication phase of the message and transmit it to the bus with a different physical layer than in the second communication phase of the message.

11. The device as recited in claim 7, wherein the additional data include at least one form of data of the following data: (i) data for signaling to the transmit/receive device an operating mode into which the transmit/receive device has to switch for a following communication phase, and/or (ii) data for configuration of the transmit/receive device, and/or (iii) data with a request to the transmit/receive device, and/or (iv) data for signaling to the transmit/receive device a differential voltage with which the transmit/receive device is to operate for the bus, and/or (v) data for programming or flashing devices, and/or (vi) data with a trigger for a predetermined wake-up pattern for a predetermined subscriber station of the bus system.

12. The device as recited in claim 7, wherein the first communication phase of the message is provided in order to negotiate which of the subscriber stations of the bus system will receive at least temporarily exclusive, collision-free access to the bus in a following second communication phase.

13. A transmit/receive device for a subscriber station of a serial bus system, the transmit/receive device comprising: a transmit module configured to transmit a transmit signal on a bus of the bus system, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between subscriber stations of the bus system; and an interface module configured to decode the transmit signal in at least one part of the second communication phase of a message into a decoded transmit signal, and to extract additional data from at least one symbol of the encoded transmit signal; wherein: the interface module is configured to extract the additional data from at least one symbol of the encoded transmit signal, which data have been inserted into the encoded transmit signal serially with information of the symbol of the transmit signal; a symbol duration of the transmit signal and of the decoded transmit signal is identical; the interface module is configured to subdivide the symbol duration of the encoded transmit signal into n symbols for a first symbol of the encoded transmit signal and into n symbols for a second symbol of the encoded transmit signal; n is a natural number greater than or equal to 1.

14. The transmit/receive device as recited in claim 13, wherein the interface module is configured to extract the information of the symbol of the transmit signal from the first or last symbol of the $2*n$ symbols of the symbol duration of the encoded transmit signal.

15. The transmit/receive device as recited in claim 13, wherein the interface module is configured to extract the additional data between the first and last symbols of the $2 \times n$ symbols of the symbol duration of the encoded transmit signal.

16. A bus system, comprising: a bus; and a plurality of subscriber stations which are connected together via the bus, wherein the bus system is configured to use a first communication phase and a second communication phase for exchanging messages between the plurality of subscriber stations of the bus system and at least one of the plurality of subscriber stations has a transmit/receive device including: a transmit module configured to transmit, to another one of the subscriber stations and via the bus of the bus system, data of one of the messages, the data having been encoded, from a transmit signal, with at least one symbol in an encoded transmit signal; and an interface module configured to decode the encoded transmit signal in at least one part of the second communication phase of the one of the messages to extract additional data from the at least one symbol of the encoded transmit signal; wherein: the additional data had been inserted into the at least one symbol of the encoded transmit signal to be interleaved serially with the data of the at least one symbol of the one of the messages; a behavior of the transmit module is controlled according to the extracted additional data; and a symbol duration of the transmit signal and of the encoded transmit signal is identical.

17. A method for a subscriber station of a serial bus system that includes a plurality of subscriber stations, the subscriber station being one of the plurality of subscriber stations, wherein the bus system is configured to use at least one first communication phase and one second communication phase for exchanging messages between the plurality of subscriber stations of the bus system, and wherein the method comprises the following steps: generating, with a communication control module, a transmit signal for a transceiver of the subscriber station, wherein the transmit signal includes data of one of the messages to be transmitted by the transceiver; encoding, using an interface module, the transmit signal in at least one part of the second communication phase of the one of the messages into an encoded transmit signal that includes at least one symbol; and inserting, with the interface module, additional data, that controls a behavior of the transceiver, into the at least one symbol of the encoded transmit signal, such that the additional data is interleaved serially with the data of the at least one symbol of the one of the messages of the transmit signal, wherein a symbol duration of the transmit signal and of the encoded transmit signal is identical.
